

VIT Bhopal University
B.Tech – Computer Science & Engineering (AI & ML)

The Digital Divide Within:

From Basic Tools to Responsible Digital Citizenship

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SECTION A

Establishing the Context

The Gap Between Using Tools and Understanding Systems

There is a quiet assumption that runs through modern student life: that because we use digital tools every day, we must understand them. We create presentations on Canva, collaborate on Google Docs, build profiles on LinkedIn and GitHub, and participate in coding communities. From the outside, this looks like fluency. But fluency in using something is not the same as understanding how it works, what it means, or what it costs.

The gap between using digital tools and truly understanding digital systems is not a minor technical distinction. It is a foundational one. A student who applies a template in Canva may not realise that the image inside it could be copyrighted. Someone who posts code on GitHub may not think about what it signals if that code is incomplete or copied. A person who clicks a link in an email may know, in the abstract, that phishing exists — and still fall for it. In each case, the tool was used. The understanding was missing.

This gap matters because digital environments are not neutral. They record behaviour, shape perception, and carry real consequences. A poorly-worded email to a faculty member does not disappear after being read. A GitHub profile that shows shallow work is still visible to a recruiter years later. A form submitted to a fraudulent website triggers a chain of events that cannot be undone simply by changing a password. The digital world does not offer the same forgiveness that casual, in-person mistakes sometimes do.

Why Widespread Platform Access Creates an Illusion of Competence

The tools most students use today are deliberately designed to lower barriers. Drag-and-drop interfaces, AI-generated suggestions, pre-built templates, and one-click collaboration features mean that a person with almost no technical background can produce output that looks sophisticated. This is a genuine achievement of modern software design — but it has an unintended consequence.

When producing a polished-looking result requires almost no deep knowledge, it becomes easy to mistake the result for the skill. A student who completes five Kaggle challenges using tutorial notebooks has done something — but may not have developed the reasoning ability to approach a new problem independently. A LinkedIn profile listing eight certifications looks impressive, but if those certifications represent shallow exposure rather than genuine capability, the profile misrepresents more than it reveals.

The platforms themselves do not distinguish between deep expertise and surface familiarity. A GitHub repository with ten commits looks the same structurally whether the code is original and sophisticated or copied and incomplete. This means the illusion of competence is not just in the

student's head — it is built into how these systems present information. Students learn, often without realising it, that visibility matters more than depth.

Digital Participation vs. Digital Literacy

Digital participation — the act of being present, active, and visible on digital platforms — is now nearly universal among students. But participation is not literacy. Literacy implies understanding: of impact, of responsibility, of risk.

A digitally literate person does not just know how to send an email — they understand how tone affects professional relationships. They do not just know that phishing exists — they pause and verify before acting, even when urgency makes that harder. They do not just build an online profile — they think about what that profile actually communicates and whether it is accurate. They understand that digital actions create records, that those records can be accessed by people they have never met, and that the consequences of careless behaviour online can follow a person for years.

The case of Arjun, described in this study, is not unusual. It is representative. He was digitally active but not always digitally aware. He used the tools correctly in a technical sense. What was missing was the layer of reflection that turns usage into literacy. That layer is what this report seeks to examine.

SECTION B

Digital Identity and Ethics

The central question of this section is whether building a digital presence genuinely benefits students, or whether it exposes them to risks they do not fully understand. My position is this: digital presence, when built deliberately and honestly, is genuinely valuable — but the way most students currently build it creates more risk than reward. The problem is not the platforms. The problem is how students relate to them.

The Equal Opportunity Argument — And Its Limits

One of the most compelling arguments for platforms like LinkedIn and GitHub is that they democratise access to opportunity. A student from a tier-3 city with no family connections in the industry can, in theory, build a public portfolio that a recruiter in Bangalore or Hyderabad can discover. This is not a small thing. The traditional professional world was gated by geography, networks, and institutional prestige. Digital platforms theoretically bypass some of those gates.

But the ideal does not fully hold in practice. The ability to present oneself well on these platforms still requires a certain kind of digital fluency — the kind that is more accessible to students who already have advantages. Access to good internet, exposure to professional norms, time to invest in building a profile, and awareness of how recruiters actually use these platforms are not equally distributed. A student who knows someone in the industry can ask how their GitHub profile looks. A student who does not has to guess. The platform is equal; the support system behind the platform is not.

Additionally, the signal value of these platforms depends heavily on context. A GitHub profile matters enormously in software hiring and barely at all in many other fields. Students who invest heavily in building digital presence in domains that do not value it may be optimising for the wrong thing entirely.

Performance Over Authenticity

The more significant concern is what digital platforms do to the relationship between students and their own learning. LinkedIn, in particular, is a platform designed to present the best version of oneself. There is nothing inherently wrong with that. But when the platform becomes the goal rather than the record of genuine achievement, the logic inverts. Students begin to pursue things because they look good on a profile rather than because they are actually learning something.

Arjun's case illustrates this precisely. He listed skills he had only briefly touched. He posted code he had not fully understood. He was not being deliberately dishonest. He was doing what the environment rewarded: looking capable. But the gap between appearance and reality does not disappear just because it goes unnoticed. It surfaces in interviews, in technical assessments, and eventually in professional contexts where actual ability is required.

The deeper issue is that this behaviour is self-reinforcing. If everyone around a student appears to be doing the same thing, the incentive to be honest about one's actual level decreases. The competitive environment on these platforms creates pressure to match the performance of peers, even when that performance is exaggerated. The result is a collective drift toward presentation over substance.

Misrepresentation: Wrong or Just Normal?

There is a genuine debate about whether exaggerating skills or presenting incomplete work is ethically wrong or simply how professional self-presentation works. Some would argue that everyone presents their best self and that recruiters understand this. Listing a skill you have begun learning is not fraud; it is aspiration.

I do not find this argument convincing, though I understand it. The line between aspiration and misrepresentation is defined by the gap between what is claimed and what can actually be delivered. Listing Python as a skill when you can write basic loops is one thing. Listing it as a proficiency when a recruiter will expect you to debug complex systems is another. The ethical problem is not presenting yourself positively — it is creating expectations you cannot meet. That has consequences not just for the individual but for the team, the organisation, and the trust the recruiter places in the platform.

Accepting misrepresentation as a norm also lowers the baseline for everyone. If the expected behaviour is to inflate, then genuine, honest profiles become invisible by comparison. The platform stops functioning as a signal of actual capability and becomes a competition in self-promotion. That is a loss for students who are genuinely skilled but uncomfortable with performance.

Digital Permanence and Accountability

Perhaps the least-discussed dimension of digital identity is permanence. The internet does not forget. A comment posted in frustration, an incorrect answer on Stack Overflow, an early GitHub repository full of poorly structured code — these remain accessible. Unlike a classroom mistake that fades with time, a digital mistake is indexed, searchable, and potentially resurfaced years later.

Should students be held accountable for everything they post, even when they lacked awareness at the time? I think the answer has to be nuanced. Full accountability for every early mistake seems disproportionate — learning involves error, and penalising all early digital activity harshly would discourage the kind of participation that eventually builds genuine competence. But accountability for patterns of behaviour is more reasonable. A single incomplete repository is forgivable. A profile that consistently misrepresents across multiple platforms over several years is a different matter.

The practical implication for students is this: build your digital presence intentionally. What you post at 19 may be seen at 24. The answer is not to avoid digital platforms — it is to engage with them honestly, to update and correct as you grow, and to treat your online profile as a record of actual development rather than a marketing exercise.

SECTION C

Collaboration vs. Misuse

Do Digital Collaboration Platforms Enhance Learning?

The strongest argument in favour of digital collaboration platforms is also the most straightforward: they work. Google Docs, Notion, shared coding environments, and project management tools like Trello or Asana genuinely enable groups of people to coordinate their thinking in ways that were simply not possible before. Version history shows who contributed what and when. Commenting features allow asynchronous discussion without requiring everyone to be in the same place. Integration with other tools means that a document can move smoothly from draft to presentation to submission without manual reformatting.

For students who learn well through discussion and iteration, these platforms are genuinely valuable. A student who would hesitate to speak up in a physical group meeting may find it easier to contribute a comment on a shared document. A student in a different time zone can contribute without being excluded by scheduling. The friction that traditionally made group work difficult — coordinating schedules, managing physical documents, tracking changes — is significantly reduced.

The Strongest Case for Unrestricted Digital Collaboration

Making the strongest possible argument for digital collaboration means acknowledging that it does more than make group work convenient — it actually models how professional work happens. In virtually every technical field, work is collaborative. Software is built by teams using shared repositories. Research papers are co-authored. Design projects involve multiple contributors working simultaneously. Restricting students from using collaborative tools does not prepare them for environments where those tools are standard. It prepares them for a model of individual, isolated work that is increasingly rare.

Furthermore, the collaborative process itself — reading others' work, building on it, giving and receiving feedback — is a form of learning that is difficult to replicate in individual assignments. The peer-learning dimension of genuine collaboration can produce understanding that a solo assignment cannot. From this perspective, any restriction on digital collaboration risks removing a legitimate and important mode of learning.

Where I Disagree — And Why

I find the argument above compelling in principle but insufficient in practice, because it depends on a version of collaboration that does not always occur. The argument assumes that students are genuinely engaging with each other's work — reading it, questioning it, improving it. In many cases, what actually happens is a division of labour that eliminates the learning benefit entirely: one person writes, others format, and everyone submits.

This is not collaboration in any meaningful sense. It is task distribution. And the platform makes it easy to disguise task distribution as collaboration, because contribution history shows that multiple people touched the document, even if most of them made only cosmetic changes. The output looks like teamwork. The learning did not happen.

The group in Arjun's case scored well. But when asked to explain specific sections, several members could not. This is the clearest possible evidence that the assignment achieved its surface goal — a polished document — without achieving its actual goal, which was understanding. The platform enabled the former while making the latter optional.

The Boundary Between Collaboration and Academic Dishonesty

The boundary between legitimate collaboration and academic dishonesty is genuinely ambiguous, and I think institutions often underestimate this ambiguity. When does helping a peer cross into doing their work? When does referencing someone's code become copying it? These questions do not have clean answers, and the fact that they do not means that enforcement is inconsistent and often unfair.

Digital environments make this harder to enforce because individual effort is less visible. In a physical exam, you can see whether someone is looking at another person's paper. In a collaborative document, you cannot easily tell whether a student read and understood what a classmate wrote or simply accepted it as the group's output without engaging with it.

The solution is not to eliminate collaborative tools but to design assessments that cannot be completed without genuine engagement — individual reflection components, viva-style follow-ups, or process documentation that shows thinking, not just output. Ethical behaviour is easier to sustain when the environment makes it easier to behave ethically than to avoid accountability.

SECTION D

Cyber Awareness and Responsibility

Why Cyber Crime Keeps Succeeding Despite Widespread Awareness

This is the question that most cyber security education fails to answer honestly. Most awareness campaigns explain what phishing is, what smishing is, what social engineering looks like. They provide checklists. They show examples. And yet, the rates of people falling for these attacks remain high — including among people who have received awareness training. Arjun had been taught about phishing. He fell for it anyway.

The reason is that cyber crime does not primarily exploit ignorance. It exploits psychology. Effective phishing attacks create urgency, simulate legitimacy, and target moments when people are emotionally engaged — excited about an opportunity, worried about a deadline, or simply tired and moving quickly. In those moments, the part of the brain that generates caution is overridden by the part that wants to act.

Arjun was excited about the internship. That emotional state made him more vulnerable, not because he forgot what phishing was, but because the desire to act quickly was stronger than the habit of verifying. This is not a failure of knowledge. It is a failure of behavioural habit — and those are much harder to change than knowledge.

User Responsibility vs. Platform and Institutional Accountability

The conventional framing of cyber safety places most of the responsibility on individual users. Do not click unknown links. Verify the sender. Use strong passwords. Enable two-factor authentication. These are reasonable practices and I am not suggesting they should be abandoned. But placing full responsibility on individual users ignores the structural environment in which those users operate.

Email platforms that allow fraudulent senders to spoof legitimate-looking addresses are part of the problem. Social media platforms that do not clearly distinguish verified accounts from impostors are part of the problem. Educational institutions that teach cyber security as a once-per-semester lecture rather than as a practised, scenario-based skill are part of the problem. Expecting individual vigilance to compensate for systemic failures is not a realistic or fair standard.

Platforms have both the technical capability and the commercial incentive to reduce phishing exposure. Better sender verification, clearer warnings for suspicious links, and real-time alerts for unusual account activity are all possible. The question of whether they are prioritised is partly a policy question and partly a market one. Until platforms are held to higher standards — through regulation or user pressure — the burden will continue to fall disproportionately on individual users who are not equipped to carry it.

Negligence, Education, or Systemic Failure?

When a student falls victim to an online scam, the easy response is to call it negligence. They should have known better. But I think this framing is both inaccurate and unhelpful. Negligence implies a failure of care that a reasonable person would not have made. Given the sophistication of modern phishing attempts and the psychological techniques they employ, falling for one is not unreasonable. It is human.

Calling it purely a lack of education is also insufficient. As I have argued above, Arjun was educated. What he lacked was not knowledge but the behavioural habit of applying that knowledge under pressure. Education can transfer knowledge. Developing habits requires repeated practice in realistic conditions — exactly the kind of scenario-based training that most institutions do not provide.

The most accurate framing is systemic failure. The system — educational, platform, regulatory — has not yet developed adequate responses to the scale and sophistication of cyber threats. Blaming individual victims obscures this and reduces pressure on the institutions and companies that have more power to change the environment than any individual user does.

Are Current Preventive Measures Sufficient?

Current preventive measures — password hygiene, two-factor authentication, awareness campaigns — address the symptoms of cyber insecurity rather than its causes. They are necessary but not sufficient. The deeper behavioural issue, the gap between knowing and doing, is rarely addressed directly.

What would actually help is cyber safety education that is scenario-based and repeated over time. Not a single module, but regular, practical exercises that simulate real attacks and give people the opportunity to practise verification habits in low-stakes environments. The military trains soldiers to act correctly under stress by drilling correct responses until they become automatic. Cyber safety education needs a similar logic: the correct behaviour must become habitual, not merely remembered.

After his experience, Arjun changed his behaviour — not because he learned something new, but because he had lived the consequence. The goal of better cyber education is to create that behavioural shift without requiring everyone to first become a victim.

SECTION E

Communication and Digital Etiquette

Why Professionalism Fails in Practice

Most students who write careless emails to faculty or professional contacts are not unaware that professional communication norms exist. They have been taught, at some point, what a formal email should look like. The problem is that this knowledge lives in a different mental compartment from their everyday digital behaviour.

Modern messaging platforms — WhatsApp, Instagram DMs, Snapchat — train us to communicate in short, fast, low-effort bursts. Speed is rewarded. Brevity is a courtesy. There is no expectation of a subject line or a formal closing. Over years of daily use, this style becomes the default — the automatic setting that activates whenever a message needs to be sent. Switching to a different register for a formal email requires conscious effort that, under time pressure or excitement, is easy to skip.

Arjun's email — 'Hi sir, I need recommendations urgently. Send asap' — reads like a WhatsApp message. It was almost certainly composed in the same mental mode. He was not being disrespectful deliberately. He was operating on autopilot in a context that required deliberate thought.

Is Informal Communication Becoming the New Normal?

There is an argument, increasingly heard in professional conversations, that formal communication norms are simply evolving. Younger professionals communicate more casually. Start-up culture has normalised first names, direct communication, and short messages. If this is the direction things are moving, perhaps insisting on formal email etiquette is training students for a world that no longer exists.

I find this argument partially true and largely overstated. It is true that communication norms vary by context — a message to a colleague at a tech start-up is appropriately different from a letter to a senior professor or a formal job application. The skill is not in knowing only one register and applying it everywhere. It is in knowing multiple registers and choosing correctly based on context. Casual communication becoming more common in some environments does not eliminate the need for formal communication in others. It increases the importance of being able to switch deliberately between them.

How Communication Style Shapes Opportunity

A poorly written email does not just fail to communicate its content effectively. It communicates something additional and unintended: that the sender did not consider the recipient's time worth the effort of a proper message. This may be entirely unfair — Arjun genuinely respected his professor — but perception is shaped by what is expressed, not by what is intended.

The stakes of this are real. A faculty member who receives a careless email may write a less enthusiastic recommendation. An employer who receives a poorly structured application may not read the attached resume carefully. A professional contact who feels addressed dismissively may not respond at all. None of these outcomes reflect the student's actual ability or character — but they follow from a single communication decision that took less than two minutes.

Digital communication is often the first and sometimes the only impression a person makes in a professional context. Treating it seriously is not about following arbitrary rules of formality. It is about recognising that how you communicate affects whether your actual message is received, and whether the relationship you are trying to build begins on a foundation of respect.

SECTION F

Personal Reflection

What I Thought Digital Literacy Meant — and What I Think Now

Before engaging with this case study, I understood digital literacy primarily as technical competence. Knowing how to use software, navigating platforms efficiently, being able to create things digitally. By that definition, most students my age — including myself — would qualify as digitally literate. We have grown up with these tools. They feel natural.

What this case study made me see more clearly is that the definition I was working from was incomplete. Using a tool well is the beginning, not the end. Digital literacy also means understanding why you are using it, what it communicates to others, what risks it carries, and what responsibilities come with it. By that fuller definition, I think most of us — myself included — have more to develop.

What Stayed With Me

The part of this case that I found most thought-provoking was not the phishing incident or the email. It was the observation about digital permanence. I had not thought carefully before about the fact that things I post now — code on GitHub, answers on forums, claims on a LinkedIn profile — remain visible and searchable for years. The idea that a recruiter in 2028 could look at something I posted in 2025 and form a judgment based on it is genuinely sobering.

It has made me think differently about what I put online — not in a way that makes me want to avoid digital platforms, but in a way that makes me want to be more intentional about them. Posting something because it looks good is different from posting something because it genuinely represents where I am and what I can do.

Where I Want to Be More Careful

If I am honest, the area where I recognise the most room to improve is professional communication. I know what a formal email should look like. But I am not certain I always pause long enough to write one when speed feels more important. This case study has reminded me that the two minutes it takes to write a proper email are rarely as expensive as the impression a careless one creates.

If I could change one thing about my current digital habits, it would be this: to pause before sending any professional communication and ask whether it reflects the effort and respect the situation deserves. That is a small change. But this case study has shown me that small communication decisions carry larger consequences than they appear to.
